Fiscal Year:	FY 2010	**Task Last Updated:**	FY 02/17/2011
PI Name:	Tash, Joseph S Ph.D.		
Project Title:	Long Term Space Flight Impacts on Male Reproductive Health		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	jtash@kumc.edu	**Fax:**	FY 913-588-7180
PI Organization Type:	UNIVERSITY	**Phone:**	913-526-4103
Organization Name:	University of Kansas Medical Center		
PI Address 1:	Department of Molecular and Integrative Physiology		
PI Address 2:	3901 Rainbow Boulevard		
PI Web Page:			
City:	Kansas City	**State:**	KS
Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	3
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:			
Flight Assignment:	STS-131		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roby, Katherine F (University of Kansas Medical Center)		
Grant/Contract No.:	NNX09AP04G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Published data demonstrates that short term microgravity (μG) (~2 wks) produces small but significant reductions in spermatogenesis in rats. However, long term ground-based μG models using the rodent hindlimb unloading (HLS) (> 3 wks), with surgical prevention of cryptorchidism, show dramatic testicular atrophy and loss of spermatogenesis leading to sterility. These results and related literature in humans suggest that longer duration space flight may have negative impacts on spermatogenesis and compromise male health and increase cancer risk. Thus, the testable hypothesis driving this proposal is: Exposure to μG 3 wks or longer causes disruption of spermatogenesis in mice. This proposal addresses the Bioastronautic Roadmap (BR) Risk 30: Chronic and Degenerative Tissue Risks in that long term μG-induced testicular atrophy will increase the risk for long term health problems post-flight including increased cancer risk and sterility. The hypothesis will be tested with the following specific aims: 1. Identify specific histopathology and dysfunction in spermatogenesis and testis function in mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 2. Quantify declines in sperm motility and morphology in cauda epididymal sperm from male mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 3. Determine dysregulation in gene transcription of markers of spermatogenic regulation and inflammatory markers in testes from mice exposed to LEO for 30d (as determined by mission duration) and compare to ground-based controls. 4. Determine declines in fertility and fecundity of male mice exposed to μG (This aim will depend upon availability of flight animals for mating trials). The opportunity to expose male mice for 30d of μG in the new BION flights represents the first to directly evaluate this reproductive health problem directly in long term μG under properly controlled conditions. These experiments will validate the ground-based long term HLS rodent model and provide proof that male reproductive health is put at risk by extended exposure to μG. The experiments will provide an initial data set for development of appropriate countermeasures to the negative impact of long term space flight on endocrine and reproductive physiology compromising male health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The observed down-regulation of estrogen receptor ERa, combined with the histological changes, suggests that the animals stopped cycling in space flight and estrogen support was severely impaired. If ERa is also down regulated in other tissues, this would indicate a common mechanism underlying space-flight induced degradation of other estrogen dependent systems including bone loss, muscle weakness, immune response, and wound repair. A broad effect of space flight on ERa receptor levels and/or signaling will offer a common pathway to generate countermeasures for space flight. Since similar changes occur in the same organ systems during the aging process, the countermeasures would have application for improving quality of life in the same organ systems during the aging process.
Task Progress:	The STS-131 BSP program was the first opportunity to determine the effects of space flight on normal adult rodent ovary/uteri. 15 adult C57Bl/6J female mice were flown on STS-131 for 15d. 16 mice were maintained in ground AEMs. Flight ovaries and uteri were harvested within 2hr of landing. The ovaries were smaller ($P<0.001$) in flight (4.04+1.53mg) compared to ground controls (5.52+0.89mg). There were significantly fewer ($P<0.002$) corpora lutea in the flight ovaries (1.57±0.61mg) vs ground (3.75±0.59mg). Most of the growing follicles in flight ovaries were atretic, indicating blocked estrous cycle. There was no significant difference in number of uterine glands, but there was a trend toward smaller uteri in flight mice. qRT-PCR revealed that expression of estrogen receptors ERa and ERb were significantly lower (>20fold) in uteri from flight compared to ground animals. Progesterone receptor Pgr-AB expression was down-regulated (3-4fold) in flight ovaries, but not uteri, while Pgr-B was lower in flight group ovaries. Lactoferrin mRNA expression is regulated by ERa and ERb levels in mice uterus and was consistently low (>20fold) during space flight. Corticosterone releasing hormone receptor CRH-R2 mRNA levels increased, though not significantly in flight mice uteri.
Bibliography Type:	Description: (Last Updated: 01/24/2022)
Abstracts for Journals and Proceedings	Tash JS, Gupta V, Holets L, Roby KF. "STS-131: Spaceflight has negative impacts on the morphology, follicle health, and steroid hormone receptors in ovaries and uterine horns in C57Bl/6J mice." To be presented at the 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. Proceedings of the 18th IAA Humans in Space Symposium, April, 2011. In press as of February 2011. , Feb-2011
Abstracts for Journals and Proceedings	Gupta V, Roby KF, Holets L, Tash JS. "Spaceflight has negative impact on the morphology and follicle health of ovaries in C57Bl/6J mice." Presented at the 26th annual meeting of the American Society for Gravitational and Space Biology, Washington, DC, November 4-7, 2010 Program and Abstracts, 26th annual meeting of the American Society for Gravitational and Space Biology, Washington, DC, November 4-7, 2010. p. 22. , Nov-2010
Abstracts for Journals and Proceedings	Holets L, Roby KF, Gupta V, Tash JS. "STS-131: Space flight induces down regulation of estrogen receptor alpha in female reproductive organs in C57Bl6 mice." Presented at the 26th annual meeting of the American Society for Gravitational and Space Biology, Washington, DC, November 4-7, 2010. Program and Abstracts, 26th annual meeting of the American Society for Gravitational and Space Biology, Washington, DC, November 4-7, 2010. p. 37. , Nov-2010
Articles in Peer-reviewed Journals	Timmerberg BD, Wolfe S, Baer LA,Tash JS. "Long term (6 week) artificial gravity at 2G does not impair reproductive health of adult male rats." Journal of Gravitational Physiology, in press 2010. Still in press, February 2011. , Feb-2011